



Christian Dimitri

Architect / BIM Developer

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Professional Summary

BIM/Software developer blending architecture and code. I build parametric and interoperable workflows across Autodesk and open-BIM stacks, focusing on real-time inspection, grading automation, and scalable data pipelines.

Skills

- Process automation & interoperability
- Parametric/Computational design
- Design optimization & QA/QC
- Frontend/Backend engineering
- APIs & Cloud (APS/Forge)
- Scrum, product thinking

Software & APIs

- AutoCAD / Civil 3D / Navisworks API; Revit API; Rhino.Inside
- Autodesk Platform Services (Forge)
- ThatOpen • Fragments & Components; IFC.js (web-ifc)
- three.js; React Three Fiber; @react-three/drei
- Rhino.Compute; Speckle Systems; Potree / E57
- WebGL / WebGPU (exp.); WebView2

Job Experience

CheckToBuild — BIM Developer (Current)

2024 – Present

- Core dev for QA/QC & inspection platform: IFC + fragments pipeline, deviation checks, and model ↔ point-cloud review.
- React Three Fiber/WebGL features using WebIFC/Fragments; performance tuning (large IFCs, E57 imagery, octree point clouds).
- Civil 3D/.NET automations (feature lines, extensions, grading utilities) with a WebView2 bridge to React.
- Data ingestion/exchange (IFC, E57, CSV/JSON), validation, and real-time overlays for site investigations.

Monstarlab — Outside Consultant, ML Auto-Grading (Civil 3D)

2025 – Present

- Own geometrical operations, data preparation, and parsing across infra for an ML MVP that auto-grades Civil 3D designs.

- End-to-end geometry pipeline: extract/clean alignments, feature lines, TIN surfaces; generate training/validation datasets.
- Topology/graph utilities (half-edge, continuity voting, dynamic angle tolerances) for robust grading line tracing.
- Contributor from day 0; internal validation currently ~99% accuracy across project scenarios.

e-verse — Software Developer (BIM)

Aug 2022 – Jul 2023

- Delivered BIM development projects on time with strong client reviews.
- Developed plugins/workflows across AutoCAD/Civil 3D, NWD, etc.
- Worked in SCRUM; built a solar-farm layout plugin to client spec.

Parametric Monkey — Software Developer (Freelance)

Mar 2022 – Jun 2022

- Transitioned a Grasshopper POC algorithm to a scalable .NET app.
- Collaborated on algorithm design and Grasshopper contributions.

Estructuras Arqué — Parametric Specialist

Aug 2020 – May 2022

- Automated structural processes with Tekla, Grasshopper, Shapediver.
- Boosted design/analysis efficiency via custom scripts and tools.

Mule Studio — Software Developer (Freelance)

Oct 2022 – Nov 2022

- Automated drawings in Tekla + Grasshopper via API integration.
- Contributed to Guggenheim UAE automation efforts.

Noumena — Computational Design Intern

Oct 2018 – Mar 2019

- Worked on digital fabrication projects with Grasshopper3D and 3D printers.
- Collaborated across disciplines to prototype novel designs.

Education

UPC — Master's in Parametric Design in Architecture (2017 – 2018)

USEK — BSc & MSc in Architectural Studies (2009 – 2016)

Releases & Projects

- Arqué Spatial Frames — Parametric App — <https://parametric-ui.netlify.app/>
- Optinave — Structural Analysis Cloud App — <http://www.optinave.com>
- Hydra — Solar Tool (Cloud) — <https://solarshapediver.e-verse.com/>
- Pris — NWD Select By Revit ID Add-in — <https://www.youtube.com/watch?v=ccjQzQ4VoaU&t=1s>
- Vero — NWD Viewpoint By Level Add-in — <https://www.youtube.com/watch?v=BXMPaEBzcwQ>